DNI Physics Program — Checkpoint 0.1
Accepted source architecture, frozen scope, and audit authorization

DATA MIND — research checkpoint; conceptual direction approved by the user

20 October 2026

Record of the accepted program
Checkpoint ID: DNI-PHYS-CP-2026-10-20-01. Status: source architecture accepted; source theories and interface not yet independently verified. Work now authorized: audit and mathematical repairs of the two source theories, not unannounced experiments or a commitment to a particular physical interpretation.

D: relativity side
The proposed deterministic-side source theory is a provenance-preserving multiarticle of Einstein's original early-20th-century special- and general-relativity works. Initial core source register:

- D-SR-1905: Albert Einstein, Zur Elektrodynamik bewegter Körper (1905), Annalen der Physik 17, 891–921. Primary special-relativity development.
- D-ME-1905: Albert Einstein, Ist die Trägheit eines Körpers von seinem Energieinhalt abhängig? (1905), Annalen der Physik 18, 639–641. Optional but relevant energy-inertia consequence.
- D-GR-1915: Albert Einstein, Die Feldgleichungen der Gravitation (25 November 1915), Sitzungsberichte der Preussischen Akademie der Wissenschaften, 844–847. Final field equations.
- D-GR-1916: Albert Einstein, Die Grundlage der allgemeinen Relativitätstheorie (1916), Annalen der Physik 49, 769–822. Extended exposition of general relativity.

Source selection policy: retain Einstein's historical text and every historical assumption with its citation; do not silently treat earlier superseded field equations as simultaneous axioms. Translation, signature, units, gauge and time-domain differences must be logged. Distinguish Einstein's original claims from later rigorous Cauchy-problem theorems. Do not claim global determinism absent appropriate data, gauge and global-hyperbolicity hypotheses.

N: quantum side
The accepted source is Asher Peres, Quantum Theory: Concepts and Methods (Kluwer, 1993), ISBN 978-0-7923-2549-9, DOI 10.1007/0-306-47120-5. N is a source-traceable probabilistic operational quantum theory, not a declaration that all quantum evolution or all ontologies are indeterministic.

The intended mathematical vocabulary includes complex Hilbert spaces, density operators, deterministic unitary evolution of isolated systems, specified quantum tests/measurements, outcome probabilities and state updates. More general completely positive quantum instruments may be added only as a separately tagged modern operational extension unless verified against the Peres source. The source is one book: its internal chapters may be audited as modules; this is not an independent multiarticle zip of multiple papers.

The mathematical DNI interface: reserved, not constructed

After separate source audits, possible mathematical relations may include typed translations, jointly interpretable structures, observation/readout maps, finite-history probabilistic comparisons, proof transformations, and incompatibility theorems. No Einstein–Peres bridge axioms are approved in this checkpoint. Neither theory may be altered merely to obtain compatibility. No quantum-gravity or determinism-versus-reality conclusion is presupposed.

Provenance and verification contract

For every record retain (source, section/page, version/translation, formal language, hypotheses, statement, proof/dependencies, checker, status). Separate historical source assertion, mathematical restatement, derived on-paper result, machine-verified result, conjecture, and repair proposal. Formal transport requires a declared proof-preserving translation; mere thematic resemblance is insufficient. No claim of full deductive closure or independent certification without receipts.

Audits now authorized

1. D: build primary-source register; compare and normalize notational variants without conflating historical theories; test special-relativistic flat sector, field equations and determinism assumptions; issue repairs and proof obligations.
2. N: map the authorized Peres book's sections; isolate source-level axioms from modern additions; check normalization, dynamics and conditional measurement calculus; issue repairs and proof obligations.
3. Keep D and N audit outputs separate; defer the actual DNI interface and all experimental runs until specifically authorized.

Baselines consulted

- The Multiarticle Theory, version 0.1 (17 Oct 2026), working research report: tagged structural zipper, low-in-fan shells and non-certification discipline.
- Determinism, Possibility, and Certified Search, manuscript 0.2 (20 Oct 2026), especially its Section 10 distinction between record unions and proof-preserving theory interfaces.

Public source locator list

- Einstein 1905 SR: https://sites.pitt.edu/~jdnorton/teaching/HPS_0410/chapters/origins_pathwa the_electrodynamics/index.html

- Einstein 1915 final field equations: https://de.wikisource.org/wiki/Die_Feldgleichungen_der_Gra
- Einstein 1916 GR: https://www.gutenberg.org/files/66944/66944-h/66944-h.htm
- Peres publisher and chapter list: https://link.springer.com/book/10.1007/0-306-47120-5
- Mathematical GR Cauchy qualification (modern, NOT an Einstein original): https://link.springer.com/article/10.1007/s41114-020-00030-z

Checkpoint limitation: full texts and all derivations have not yet been independently formalized or checker-certified. The present document preserves the research decisions and freezes the boundary before audit/repair.